Palo Alto Firewall
File Blocking

1. Lab Objective

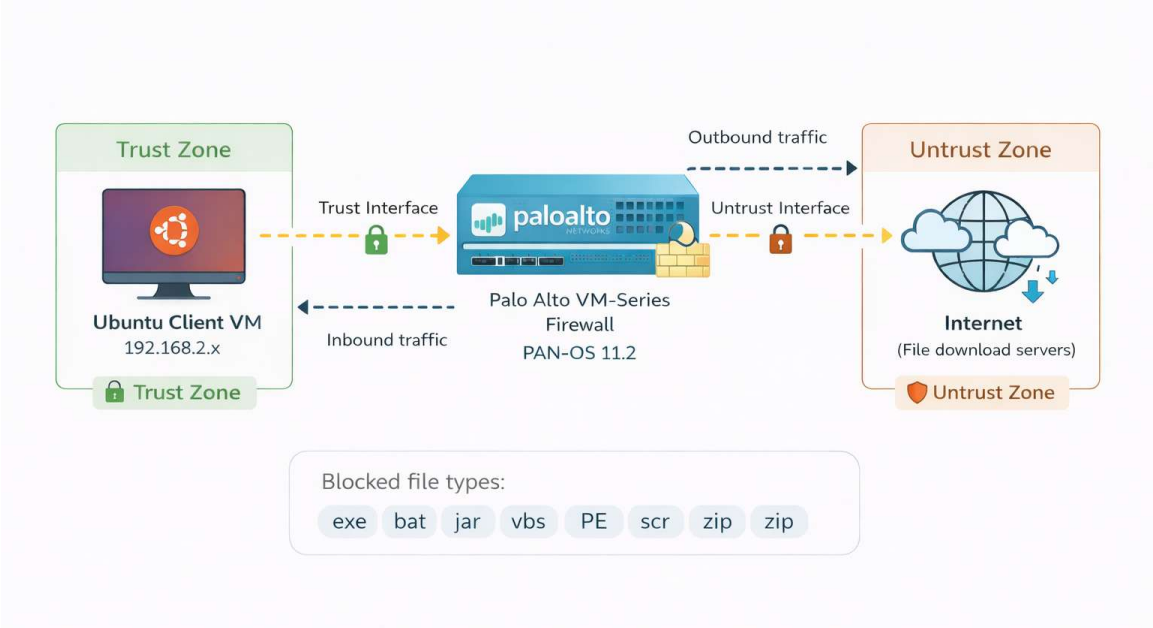
The objective of this lab is to configure a File Blocking Security Profile on the Palo Alto VM-Series Firewall to prevent users from downloading potentially dangerous file types such as executables, scripts, and archives.

Lab Environment

Firewall: Palo Alto VM-Series
PAN-OS Version: 11.2
Hypervisor: VMware Workstation
Client Machine: Ubuntu 24.04
Internet Access: Enabled via NAT

Network Topology Diagram

The below diagram represents the lab architecture where the Ubuntu VM connects to the internet through the Palo Alto Firewall.



2. Configuration Steps

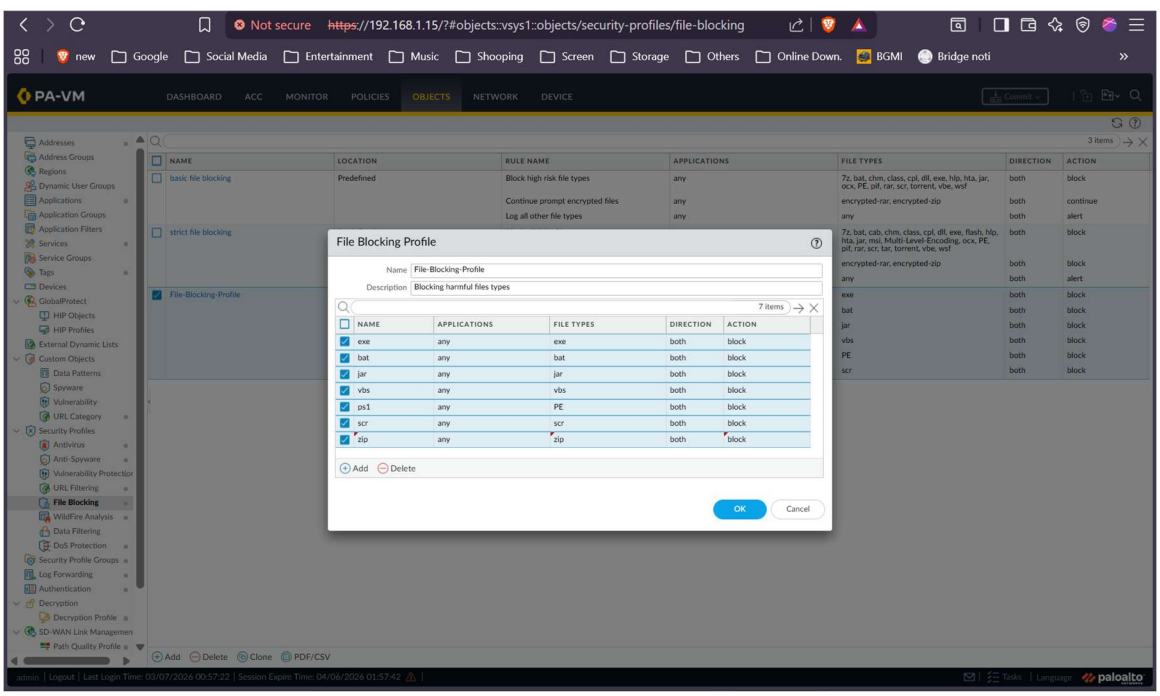
Step 1 – Create File Blocking Profile

Navigate to Objects → Security Profiles → File Blocking

Create a new profile named 'File-Blocking-Profile'. Configure rules to block high-risk file types.

Blocked file types configured in this lab:

- **exe**
- **bat**
- **jar**
- **vbs**
- **PE**
- **scr**
- **zip**

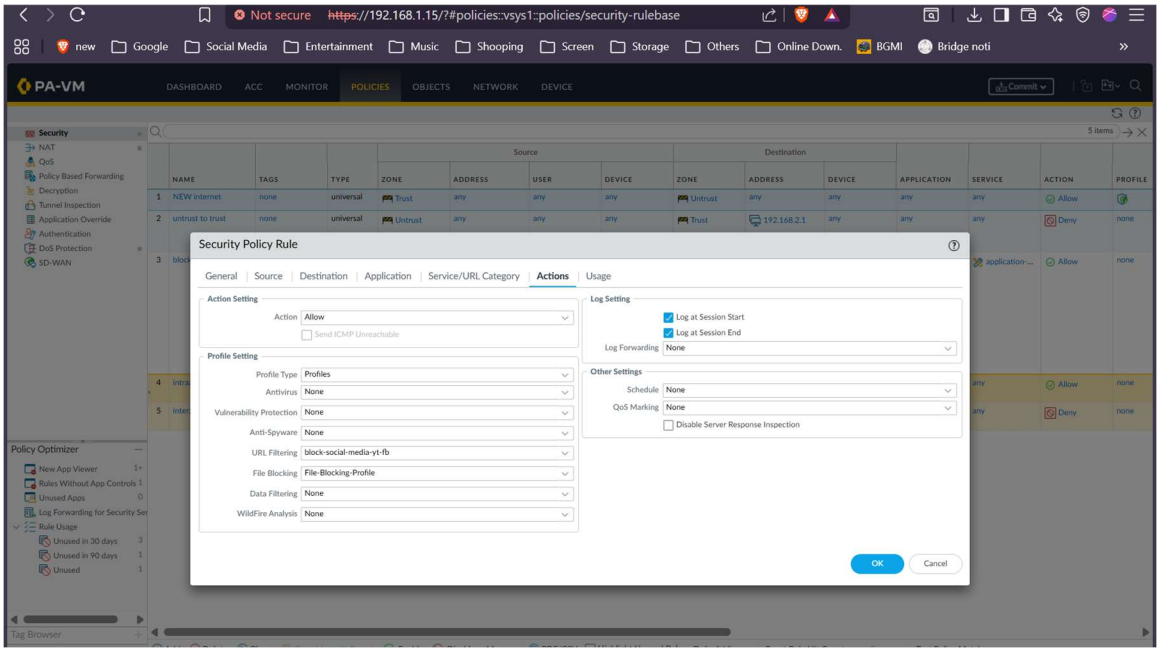


Step 2 – Attach File Blocking Profile to Security Policy

The File Blocking profile must be attached to a Security Policy rule so that traffic passing through the firewall is inspected.

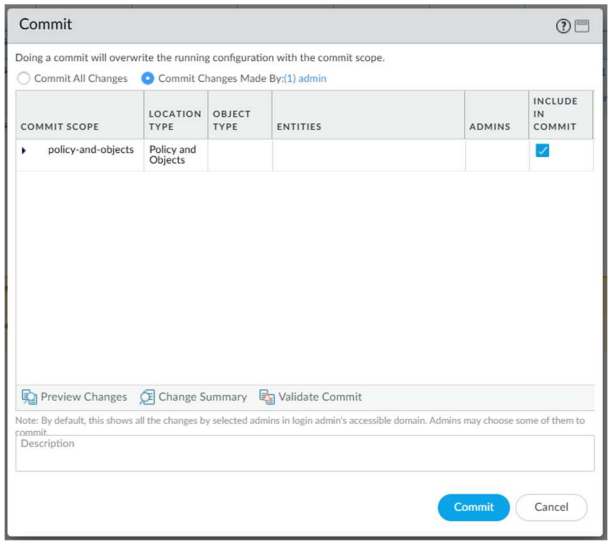
Navigate to Policies → Security and edit the rule allowing Trust → Untrust traffic.

Under the Actions tab select Profile Type = Profiles and choose the File-Blocking-Profile.



Step 3 – Commit Configuration

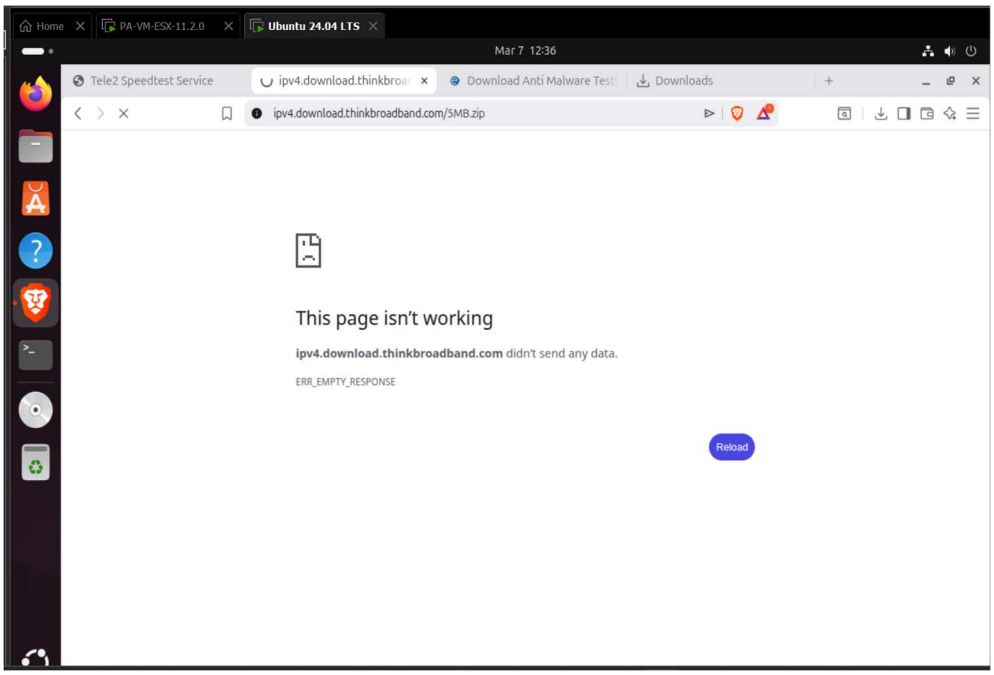
After configuring the profile and attaching it to the security policy, commit the configuration so the firewall begins enforcing the new settings.



Step 4 – Generate Test Traffic

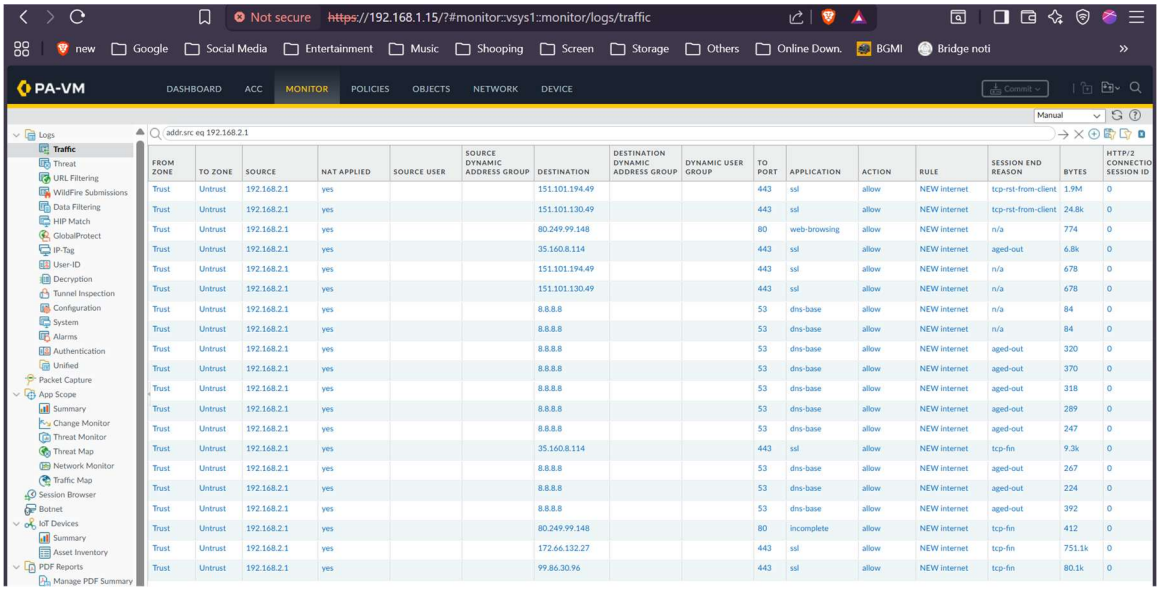
To verify File Blocking, attempt to download a ZIP file over HTTP from a public test server.

Test URL used: http://ipv4.download.thinkbroadband.com/5MB.zip



Step 5 – Verify Firewall Logs

When the download is attempted, the firewall detects the blocked file type and resets the connection. The browser shows an error such as 'ERR_EMPTY_RESPONSE', indicating the session was terminated by the firewall. Traffic logs confirm that the session matched.



FROM ZONE	TO ZONE	SOURCE	NAT APPLIED	SOURCE USER	SOURCE DYNAMIC ADDRESS GROUP	DESTINATION	DESTINATION DYNAMIC ADDRESS GROUP	DYNAMIC USER GROUP	TO PORT	APPLICATION	ACTION	RULE	SESSION END REASON	BYTES	HTTP/2 CONNECTIO SESSION ID
Trust	Untrust	192.168.2.1	yes			151.101.194.49			443	ssl	allow	NEW internet	tcp-rst-from-client	1.9M	0
Trust	Untrust	192.168.2.1	yes			151.101.130.49			443	ssl	allow	NEW internet	tcp-rst-from-client	24.8k	0
Trust	Untrust	192.168.2.1	yes			80.249.99.148			80	web-browsing	allow	NEW internet	n/a	774	0
Trust	Untrust	192.168.2.1	yes			35.160.8.114			443	ssl	allow	NEW internet	aged-out	6.8k	0
Trust	Untrust	192.168.2.1	yes			151.101.194.49			443	ssl	allow	NEW internet	n/a	678	0
Trust	Untrust	192.168.2.1	yes			151.101.130.49			443	ssl	allow	NEW internet	n/a	678	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	n/a	84	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	n/a	84	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	320	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	370	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	318	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	289	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	247	0
Trust	Untrust	192.168.2.1	yes			35.160.8.114			443	ssl	allow	NEW internet	tcp-fin	9.3k	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	267	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	224	0
Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet	aged-out	392	0
Trust	Untrust	192.168.2.1	yes			80.249.99.148			80	incomplete	allow	NEW internet	tcp-fin	412	0
Trust	Untrust	192.168.2.1	yes			172.66.132.27			443	ssl	allow	NEW internet	tcp-fin	751.1k	0
Trust	Untrust	192.168.2.1	yes			99.86.30.96			443	ssl	allow	NEW internet	tcp-fin	80.1k	0

3. Conclusion

The Palo Alto VM-Series Firewall successfully blocked the download of restricted file types using the File Blocking Security Profile. When a user attempted to download a ZIP file, the firewall terminated the session and prevented the file from being transferred.

- Vishesh Dixit